



WHAT HAPPENS WHEN

E-COMMERCE REVENUE JUMPS +137% YOY?

Behind Target's 100,000 orders across 27 Brazilian states:
An Operational & Regional Logistics SQL Deep Dive. 

99,441

Delivered Orders Analyzed

Sept 2016 – Aug 2018 span

27 States

4,119 Distinct Cities

Nationwide consumer reach

3.5x Gap

Delivery Lead Time Gap

29.3 days (RR) vs 8.3 days (SP)

+137.38%

YoY Revenue Surge

Jan–Aug 2017 vs 2018 pacing

EXECUTIVE SUMMARY & PROJECT CONTEXT:

- Solved as part of Scaler Academy's DSML Curriculum using Google BigQuery.
- Grounded in 1 year of enterprise operations analytics experience at BPCL.
- Proves how advanced SQL (CTEs, DENSE_RANK, LAG) uncovers regional supply chain bottlenecks, freight disparities, and consumer payment patterns.
- Swipe through for queries, data findings, and strategic recommendations.



THE BUSINESS PROBLEM

Scaling Across 4,119 Cities

Target experienced rapid e-commerce expansion in Brazil between 2016 and 2018. However, managing orders across a vast continent presents severe supply chain, freight cost inflation, and regional delivery lead-time challenges.

1. Regional Logistics

Are distant northern territories suffering from unacceptable fulfillment delays?

2. Freight Equity

Are high shipping fees causing cart abandonment outside southeastern hubs?

3. Revenue Velocity

Is the top-line growth pacing sustainable across matching seasonal periods?

4. Consumer Habits

When do customers order, and how heavily do they rely on installment credit?

RELATIONAL DATA ARCHITECTURE (8 TABLES IN GOOGLE BIGQUERY):

- **orders (99.4K):** Timestamps: purchase, carrier handover, actual delivery & promised SLA
- **customers (99.4K):** Customer identifiers, unique repeat keys, cities & 27 state codes
- **order_items (112.6K):** Item pricing (BRL), individual freight costs, product & seller keys
- **payments (103.8K):** Payment methods (Credit Card, Boletto) and installment sequences
- **sellers, products, reviews:** Geolocation (1M+ coordinates), product dimensions, and review scores

FINDING #1: FULFILLMENT VELOCITY

A 3.5x Regional Lead-Time Gap

Comparing average delivery lead time between best and worst performing states:

[FASTEST FULFILLMENT] SÃO PAULO (SP)

8.3 Days

Average Delivery Time from purchase to customer doorstep.
Consistently arrives 11+ days ahead of estimated SLA dates.

[SLOWEST FULFILLMENT] RORAIMA (RR)

29.3 Days

Average Delivery Time — almost an entire month waiting time!
~3.5x longer delivery cycle compared to São Paulo.

STATE-BY-STATE LOGISTICS LEAD TIME (TOP 5 EXTREMES)

State (Slowest)	Avg Delivery (Days)	State (Fastest)	Avg Delivery (Days)
RR (Roraima)	29.3 days	SP (São Paulo)	8.3 days
AP (Amapá)	27.2 days	PR (Paraná)	11.6 days
AM (Amazonas)	26.0 days	MG (Minas Gerais)	11.6 days
PA (Pará)	23.4 days	SC (Santa Catarina)	14.5 days
MA (Maranhão)	21.1 days	DF (Distrito Federal)	12.6 days

OPERATIONAL TAKEAWAY:

Northern territories suffer because fulfillment routes originate almost entirely from southeastern warehouses. Target cannot scale nationwide without local hubs.



QUERY ARCHITECTURE

Window Ranking with DENSE_RANK()

How I isolated the top 5 highest and lowest lead-time states without breaking rank sequence:



BIGQUERY SQL • DUAL DENSE_RANK() OVER STATE AVERAGES

```
-- 1. Calculate actual transit duration per delivered order
WITH dt AS (
  SELECT o.order_id, c.customer_state,
         DATE_DIFF(order_delivered_customer_date,
                   order_purchase_timestamp, DAY) AS time_to_deliver
  FROM `target_sql.orders` o
  JOIN `target_sql.customers` c USING (customer_id)
  WHERE order_delivered_customer_date IS NOT NULL
),
-- 2. Compute state-level averages and window rankings
tr AS (
  SELECT customer_state,
         ROUND(AVG(time_to_deliver), 2) AS avg_delivery_time,
         DENSE_RANK() OVER (ORDER BY AVG(time_to_deliver) DESC) AS high_rank,
         DENSE_RANK() OVER (ORDER BY AVG(time_to_deliver)) AS low_rank
  FROM dt GROUP BY customer_state
)
-- 3. Extract top 5 extremes dynamically in one query pass
SELECT customer_state, avg_delivery_time,
       CASE WHEN high_rank <= 5 THEN 'top 5 highest'
            WHEN low_rank <= 5 THEN 'top 5 lowest' END AS tier
FROM tr WHERE high_rank <= 5 OR low_rank <= 5
ORDER BY avg_delivery_time DESC;
```

WHY DENSE_RANK() INSTEAD OF ROW_NUMBER() OR RANK()?

- Handles Ties Cleanly: States with identical average delivery times share the same rank.
- No Skipped Positions: Consecutive ranks remain intact (1, 2, 3... rather than 1, 1, 3).
- Dual Window Strategy: Generates both high and low rankings in a single database scan.



FINDING #2: FREIGHT DISPARITY

The Hidden Penalty on Remote States

Customers in distant territories don't just wait longer—they pay ~2.8x more for freight:

SP (São Paulo)

17.37 BRL

Lowest Avg Freight across Brazil

RR (Roraima)

48.59 BRL

2.8x higher than São Paulo!

WHY FREIGHT DISPARITY CAUSES CHECKOUT ABANDONMENT:

Seller Clustering in Southeast:

Over 70% of active sellers and warehouses are situated in SP, PR, and RJ. Long-haul overland transport drives up freight rates.

The 'Distance Tax':

For a typical 100 BRL basket, an extra 48 BRL freight charge represents a nearly 50% checkout surcharge on the consumer!

Depressed Northern Orders:

States like RR and AP have order volumes barely reaching double digits—not from low desire, but because shipping cost is prohibitive.

THE STRATEGIC INTERVENTION:

Target cannot rely purely on digital marketing in northern Brazil.

Marketing spend is wasted if customers drop out at payment due to freight.

Subsidized shipping tiers or regional seller onboarding is necessary to unlock demand.

FINDING #3: ORDER BEHAVIOR

When & How Brazilian Consumers Buy

Analyzing ordering hours and payment preferences reveals two vital operational patterns:

1. PEAK ORDERING WINDOW: AFTERNOON (13:00 – 18:00)

Afternoon (13:00 - 18:00):	38,135 Orders	— PEAK TRAFFIC (Weekday avg 14:00 - 15:00)
Night (19:00 - 23:00):	33,518 Orders	— High evening transaction activity
Morning (07:00 - 12:00):	23,359 Orders	— Gradual ramp-up during workday
Dawn (00:00 - 06:00):	4,429 Orders	— Minimal activity window (ideal for maintenance)

Operational Action: Schedule BigQuery ETL and platform database backups

strictly during Dawn (02:00 - 05:00) to ensure zero friction during afternoon peaks.

2. PAYMENT METHODS & MULTI-INSTALLMENT RELIANCE

Credit Card is dominant (>75% of orders), followed by Boleto bancário.

• 1 Installment:	52,580 orders (50.6%)	— Dominates low-to-mid ticket items
• 2 – 4 Installments:	27,713 orders (26.7%)	— Common for fashion and home essentials
• 5 – 10 Installments:	22,544 orders (21.7%)	— Crucial for electronics and high-AOV items
• > 10 Installments:	1,049 orders (1.0%)	— High-ticket special purchases

Commercial Insight: Nearly 50% of orders rely on installment financing.

Promoting zero-fee EMI / installment discounts directly boosts checkout conversions.



QUERY ARCHITECTURE

Measuring YoY Growth with LAG()

Comparing identical 8-month windows (Jan–Aug) in 2017 vs 2018 to eliminate seasonality bias:



BIGQUERY SQL • FINANCIAL REVENUE PACING VIA LAG()

```
-- 1. Aggregate total revenue for Jan-Aug in 2017 and 2018
WITH tcs AS (
  SELECT
    EXTRACT(YEAR FROM o.order_purchase_timestamp) AS order_year,
    ROUND(SUM(p.payment_value), 2) AS order_total_cost
  FROM `target_sql.orders` o
  JOIN `target_sql.payments` p ON o.order_id = p.order_id
  WHERE EXTRACT(YEAR FROM o.order_purchase_timestamp) IN (2017, 2018)
    AND EXTRACT(MONTH FROM o.order_purchase_timestamp) BETWEEN 1 AND 8
  GROUP BY EXTRACT(YEAR FROM o.order_purchase_timestamp)
),
-- 2. Window LAG to access prior year revenue in the same row
lags AS (
  SELECT order_year, order_total_cost,
    LAG(order_total_cost) OVER (ORDER BY order_year) AS pre_year_cost
  FROM tcs
)
-- 3. Compute exact percentage growth rate
SELECT order_year, order_total_cost, pre_year_cost,
  ROUND((order_total_cost - pre_year_cost)/pre_year_cost * 100, 2)
  AS percentage_growth
FROM lags ORDER BY order_year;
```

RESULT SET FROM BIGQUERY EXECUTION:

- 2017 (Jan–Aug): 3,707,750.32 BRL
- 2018 (Jan–Aug): 8,801,634.64 BRL

VERIFIED YoY REVENUE GROWTH: +137.38%



BUSINESS RECOMMENDATIONS

4 Data-Backed Action Plans

Translating data insights into high-ROI operational and supply chain decisions:

1. Establish Northern Regional Micro-Hubs

Partner with localized 3PL providers in Manaus or Belém to slash the 29-day delivery cycle down to under 12 days, reducing cancellation and churn rates.

2. Implement Tiered Freight Subsidies

Introduce free shipping thresholds (e.g., Orders > 150 BRL) for remote states to eliminate the 48 BRL checkout barrier and unlock dormant customer demand.

3. Provision Infrastructure for Afternoon Peaks

Autoscale database resources and customer support between 13:00 and 18:00 when ~40% of transactions occur; shift system backups to 03:00 dawn.

4. Expand Zero-Cost Installment Programs

With ~50% of orders relying on installment financing, offer zero-fee EMI promotions on high-AOV categories to increase average cart size.



CAPABILITY SHOWCASE

Skills & Technologies Applied

How my analytical and operational background was leveraged in this case study:

Google BigQuery

Cloud Relational Querying

Multi-table relational schema exploration, column validation, large table joins across 100K+ records.

CTEs, DENSE_RANK(), LAG()

Advanced SQL Patterns

Common Table Expressions, window ranking for state extremes, lag pacing for YoY financial metrics.

Delivery SLA & Freight Modeling

Supply Chain & Logistics

Measuring actual vs estimated transit days, identifying shipping cost inflation by territory.

Ex-BPCL Operations Grounding

Applied Operational Acumen

1 year monitoring retail sales & stock across 350+ petrol stations applied directly to retail logistics.

DIVE DEEPER

Explore the Code & Case Study

Everything in this project is open-source and documented in detail:



GitHub Repository

github.com/Hazardous9hub/Target-SQL-Business-Case-Study

- Complete SQL scripts, 8-table schema docs, and 39 BigQuery query execution screenshots.

Live Portfolio Website

iamshivalingbattarki09.vercel.app

- Interactive case studies, live project demos, verified certifications, and downloadable resume.



LinkedIn Profile

linkedin.com/in/shivaling-93000

- Connect with me for data analytics discussions, operational problem solving, and opportunities.



SHIVALING BATTARKI

Data Analyst | Ex-BPCL | Scaler DSML Fellow | B.E. Mechanical (8.67 CGPA)

- 1 Year on-ground operations analytics at BPCL (monitoring 350+ petrol stations & SAP ECC).
- HackerRank 5-Star Gold SQL | Scaler DSML Fellow | Google BigQuery & Tableau Specialist.

Actively seeking Data Analyst & Business Analyst roles. Let's connect!